

ON SEPARATING SEMIGROUPS OF (M-2)-CURVES

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1. Preliminaries. A *real algebraic curve* X is a complex algebraic curve endowed with an antiholomorphic involution $\text{conj}: X \rightarrow X$. The set of real points of X , denoted by $\mathbb{R}X$, is the fixed point set of the involution conj . All curves considered here are projective, smooth and irreducible.

A real curve X is called *separating* or *dividing* if $X \setminus \mathbb{R}X$ is disconnected. A classical result of Ahlfors states that a real curve X is separating if and only if it admits a *separating morphism*. That is, a real morphism $f: X \rightarrow \mathbb{P}^1$ such that $f^{-1}(\mathbb{R}\mathbb{P}^1) = \mathbb{R}X$.

Every separating morphism defines a covering map $f|_{\mathbb{R}X}: \mathbb{R}X \rightarrow \mathbb{R}\mathbb{P}^1$. Fix a numbering of the connected components $X_1, \dots, X_r$ of $\mathbb{R}X$. Denoting by $d_i(f)$ the degree of the restriction of f to X_i , we can assign to each separating morphism f the tuple of positive integers $d(f) = (d_1(f), \dots, d_r(f))$. Kummer and Shaw defined the *separating semigroup* of X as the set of all such tuples:

$$\text{Sep}(X) := \{d(f) \mid f: X \rightarrow \mathbb{P}^1 \text{ is a separating morphism}\}.$$

One can easily check that $\text{Sep}(X)$ is indeed a semigroup; see [KS20, Prop. 2.1]. We denote $\mathbb{N} := \{n \in \mathbb{Z} \mid n \geq 1\}$ and $\mathbb{N}_0 := \{n \in \mathbb{Z} \mid n \geq 0\}$.

By the Harnack inequality, the number of connected components of the real locus of a real curve of genus g does not exceed $g + 1$. A real curve X is called an M -*curve* if this number is maximal, i.e., $b_0(\mathbb{R}X) = g + 1$. If $b_0(\mathbb{R}X) = g + 1 - i$, then X is called an $(M - i)$ -*curve*. The separating semigroups of M -curves are easily computed: it is shown in [KS20] that $\text{Sep}(X) = \mathbb{N}^{g+1}$ for such curves. By Klein's congruence, for a separating $(M - i)$ -curve X we have $b_0(\mathbb{R}X) \equiv g + 1 \pmod{2}$, so i must be even. Consequently, after M -curves, the next natural case to consider is that of $(M - 2)$ -curves. The present note is devoted to describing their separating semigroups.

The separating semigroups have already been computed for $(M - 2)$ -curves of small genus. If $g = 3$, then $\text{Sep}(X) = (2, 1) + \mathbb{N}_0^2$ (see [Ore19]). If $g = 4$, then, as shown in [Ore25], $\text{Sep}(X)$ is either $\mathbb{N}^3$, $(2, 1, 1) + \mathbb{N}_0^3$, or $\{(1, 1, 1)\} \cup ((1, 2, 1) + \mathbb{N}_0^3)$. If X is a plane quintic (in which case $g = 6$), it is shown in [MO25a] that $\text{Sep}(X) = (2, 1, 1, 1, 1) + \mathbb{N}_0^5$.

The description of the separating semigroup depends crucially on the minimal degree of a separating morphism. We denote this value by $\text{sepgon}(X)$ and call it the *separating gonality* of X . It follows from results of A. Gabard [Gab06] that for an $(M - 2)$ -curve X of genus g , we have $g - 1 \leq \text{sepgon}(X) \leq g$, so only two cases are possible: $\text{sepgon}(X) = g - 1$ or $\text{sepgon}(X) = g$.

In [Man26], M. Manzaroli studies the connection between separating morphisms and totally real pencils of curves of degree $d - 3$ on plane separating $(M - 2)$ -curves of degree d . In addition, the paper contains partial results on the separating semigroups of $(M - 2)$ -curves of arbitrary genus. Specifically, it is shown that $(3, \dots, 3) + \mathbb{N}_0^{g-1} \subset \text{Sep}(X)$ if $\text{sepgon}(X) = g - 1$, and $(4, 2, \dots, 2) + \mathbb{N}_0^{g-1} \subset \text{Sep}(X)$ if $\text{sepgon}(X) = g$.

In the present paper, building on recent results by G. Mikhalkin and S. Orevkov, we refine these observations. We call a divisor $P = p_1 + \dots + p_n$ on X *separating* if it is a fiber of some separating morphism; i.e., if there exists a separating morphism $f: X \rightarrow \mathbb{P}^1$ such that $P = f^{-1}(p_0)$ for some $p_0 \in \mathbb{R}\mathbb{P}^1$. In this case, we write $d(P) := d(f)$.

The following theorem was proved in [MO25b].

Theorem 1 ([MO25b]). Let X be a separating real curve of genus g with $b_0(\mathbb{R}X) = r$.

- (a) Let $P = p_1 + \dots + p_g$ be a separating divisor on X with $h^1(P) = 1$. Then $d(P) + \mathbb{N}_0^r \subseteq \text{Sep}(X)$.
- (b) Assume that X is not hyperelliptic. Let $Q = q_1 + \dots + q_{g-1}$ be a separating divisor with $h^0(Q) = 2$ such that Q is not a theta characteristic (i.e., $2Q \not\sim K_X$). Then $d(Q) + \mathbb{N}_0^r \subseteq \text{Sep}(X)$.

This theorem allows us to deduce much about the separating semigroups of $(M-2)$ -curves in the general case. To formulate the main result, we need the following terminology.

Following Coppens [Cop14], we say that a separating $(M-2)$ -curve X of genus g is *of special type* if there exists a connected component $X' \subset \mathbb{R}X$ with the following property: for every degree g morphism $f: X \rightarrow \mathbb{P}^1$ which has odd degree on every connected component $X'' \neq X'$ we have $f(X') = \mathbb{R}\mathbb{P}^1$. Otherwise, X is said to be *not of special type*. It is straightforward to see that X is a curve of special type if every morphism $f: X \rightarrow \mathbb{P}^1$ of degree g whose restriction to each connected component of $\mathbb{R}X$ except one has odd degree is separating (in this case, its topological degree is $(2, 1, \dots, 1)$). Conversely, X is of non-special type if there exists a non-separating morphism of degree g from X to $\mathbb{P}^1$ that has odd degree on all connected components of $\mathbb{R}X$ except one (in this case, its topological degree is $(0, 1, \dots, 1)$).

2. Main result. The main goal of this note is to prove the following theorem.

Theorem 2. Let X be a separating $(M-2)$ -curve of genus g .

- (1) Assume that $\text{sepgon}(X) = g$ and let f be a separating morphism of degree g . Number the connected components of $\mathbb{R}X$ so that $d_1(f) = 2$. Then

$$(2, 1, \dots, 1) + \mathbb{N}_0^{g-1} \subseteq \text{Sep}(X).$$

- (2) Assume that $\text{sepgon}(X) = g - 1$, $g \geq 5$, and X is general¹ and not of special type. Then

$$\text{Sep}(X) = \mathbb{N}^{g-1}.$$

Proof. (1) Let P be a separating divisor of degree g corresponding to the separating morphism f . Note that the divisor $K_X - P$ has at most one point on each connected component of $\mathbb{R}X$. Thus, by Corollary 2.2 of [Hui01], we have $h^0(K_X - P) = 1$. Thus, the claim follows immediately from part (a) of Theorem 1.

(2) We apply part (b) of Theorem 1. Since the number of connected components of the real locus of a separating hyperelliptic curve of genus g can be 1, 2, or $g+1$ (see [GH81, Prop. 6.3]), a real $(M-2)$ -curve cannot be hyperelliptic for $g \geq 4$.

We now show that a separating divisor of degree $g-1$ on X can be chosen so that it is not a theta characteristic. Let $W_d^1(X)$ denote the subvariety of $\text{Pic}^d(X)$ parametrizing complete linear systems of degree d and rank at least 1. Consider the subvariety $\mathcal{S}_d \subset W_d^1(X)$ consisting of separating linear systems of degree d , i.e.,

$$\mathcal{S}_d := \{|D| \mid \deg D = d, D \text{ is separating}\} \subset W_d^1(X).$$

As noted in [Cop14], $\mathcal{S}_d$ is a dense open subset of some irreducible component of $\mathbb{R}W_d^1(X)$. It follows from Brill–Noether theory (see [PG80]) that for a general curve of genus g and $d \leq g+1$, the equality $\dim W_d^1(X) = 2d - g - 2$ holds. In particular, $\dim W_{g-1}^1(X) = g - 4$. Since X is a general $(M-2)$ -curve of non-special type, all irreducible components of $\mathbb{R}W_{g-1}^1(X)$ have real dimension $g-4$ (as noted in [Cop14]). Consequently, since $\mathcal{S}_{g-1}$ is a dense open subset of an irreducible component of $\mathbb{R}W_{g-1}^1(X)$, we obtain $\dim \mathcal{S}_{g-1} = g - 4$.

Therefore, since $\mathcal{S}_{g-1}$ is at least one-dimensional and there are only finitely many² theta characteristics on X , we can choose a separating divisor of degree $g-1$ that is not a theta characteristic.

Now, let $Q = q_1 + \dots + q_{g-1}$ be a separating divisor such that $2Q \not\sim K_X$. Again, by Corollary 2.2 of [Hui01], we have $h^0(Q - q_i) = 1$, hence $h^0(Q) \leq 2$. On the other hand, since Q is separating, we have $h^0(Q) \geq 2$, which implies $h^0(Q) = 2$. The claim then follows from part (b) of Theorem 1. $\square$

Let Teich_g^{M-2} denote the Teichmüller space of separating $(M-2)$ -curves of genus g . Let $\text{Teich}_{g,s}^{M-2}$ and $\text{Teich}_{g,ns}^{M-2}$ denote the subsets of curves of special and non-special type, respectively. Coppens [Cop14, Lemma 2.6] showed that $\text{Teich}_{g,ns}^{M-2}$ is an open subset of Teich_g^{M-2} . Combining this with part (2) of Theorem 2, we obtain the following corollary:

Corollary 3. Let $g \geq 5$. For curves in an open subset of Teich_g^{M-2} , the separating semigroup equals $\mathbb{N}^{g-1}$.

¹Note that a general separating $(M-2)$ -curve can be either of special type or not; see [Cop14, Corollary 4.8].

²By definition, any two theta characteristics differ by an element of the 2-torsion subgroup of the Jacobian of X . Since this subgroup is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^{2g}$, there are exactly 2^{2g} pairwise inequivalent theta characteristics on X .

Remark 1. For $(M-2)$ -curves of special type, however, the argument used in the proof of part (b) fails. Coppens [Cop14, Prop. 4.1] showed that for $g \geq 3$, any linear system of degree $g-1$ and rank 1 having degree 1 on each connected component is half-canonical. Consequently, part (b) of Theorem 1 cannot be applied.

Remark 2. As mentioned in the introduction, in all cases described so far, the separating semigroup of an $(M-2)$ -curve of genus g with separating gonality g equals $(2, 1, \dots, 1) + \mathbb{N}_0^{g-1}$ for a suitable numbering of the connected components of $\mathbb{R}X$. It is therefore natural to conjecture that in part (1) of Theorem 2, equality actually holds rather than just inclusion.

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